

3(a)	any three from: to award MPs 2, 3 and 6 there must be reference to shape or conformation at least once in the answer if no reference to shape or conformation in the answer mark to max 2 if lock and key described mark to max 2 from MP4 onwards 1 <i>idea that</i> active site (of inactive enzyme) and substrate are, (only) partially / not, complementary ; 2 as substrate enters active site there is a, change in shape / conformational change, (of active site / enzyme) ; A active site moulds around substrate R substrate changes shape / substrate active site 3 so shape of active site becomes complementary to substrate ; A <i>idea that</i> there is a good fit between active site and substrate 4 enzyme-substrate complex forms ; 5 lowers activation energy ; 6 product(s) leave and, enzyme / active site, returns to original, shape / conformation ; 7 AVP ; e.g. catalytic sites move to be in correct position e.g. detail of lowering activation energy – putting strain on substrate / stress on bonds	3
3(b)(i)	allow enzyme or subtilisin for immobilised subtilisin A allow attachment for attachment of larvae I comparisons between controls any four from: 1 effective because (much) lower percentage attachment with, the enzyme / AW, compared with the control(s) / AW ; ora for (both) controls 2 any comparison between 24 and 48 hrs in the same experimental tank (1 to 4) ; e.g. tank 2 only one showing decrease at 48 hours 3 any comparison between any two different tanks (1 to 4 or any experimental with control) at either 24 or 48 hours ; e.g. less attachment in tank 2 than in the others at 48 hours 4 48 hours is a short period of time / may not be effective after a few days ; 5 not 100% effective ; 6 (generally) becomes less effective at 48 hours (with exception of tank 2) ; 7 immobilised enzyme, not stable / detached from surface ; 8 <i>ref. to</i> no trend with increasing subtilisin concentration ; 9 no repeats so data may not be valid / AW ; 10 AVP ;	4

Question	Answer	Marks
3(b)(ii)	<i>I ref. to cost</i> any two from: 1 density / concentration of, enzyme / subtilisin A / polymer ; A way of attaching enzyme to polymer 2 type of polymer ; 3 check that all organisms are removed from ships before applying, enzyme / polymer ; 4 way to fix polymer to surfaces of ships / material used to make surface of ships ; 5 number of larvae put into tanks / experiment carried out at time of year when larvae are in sea water ; A idea that different places have different numbers of larvae 6 length of time before taking measurements ; 7 length of time, enzyme / anti-fouling agent, remains active ; A ref. to stability of immobilised enzyme 8 <i>idea</i> of any interaction with other, fouling / marine, organisms ; A 'affect' as an interaction e.g. that could, graze / decompose, subtilisin A / polymer that could compete with <i>A. amphitrite</i> for attachment sites effectiveness against fouling organisms other than, <i>A. amphitrite</i> / barnacles subtilisin A / product of enzyme, may be, toxic / harmful, to marine organisms 9 effect of immersion in sea water on, enzyme / polymer (not in artificial sea water) ; 10 effect of different types of sea water, e.g. salt content / mineral content / roughness ; A different, concentrations of sea water / salinities 11 effect of marine pollutants / something in sea water, as enzyme inhibitors ; 12 AVP ; e.g. <i>ref.</i> to surface area of ship exposed to sea water types of antifouling agents present in the seawater	2